Impression weighted memory design

Part 3 The compression train

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Rick's addition to Part 2 is to assess a heavily compacted "cart" together with the carts ahead and behind it, rather than requiring each cart to reconstruct itself alone. This can support lossless reconstruction when the retained bundle and its decoding rules preserve all the information needed to identify the original exactly. It is a conditional design proposal, not a demonstrated memory architecture.

A cart need not carry everything alone

A cart represents a record or block. Other records can supply side information: a shared phrase, a dictionary entry, a reference version or a predictable pattern. The current cart can then store a reference and the differences, instead of repeating everything. An isolated cart may be insufficient even though the complete, correctly linked bundle is lossless.

This principle has established technical counterparts. DEFLATE uses references to matching earlier data; VCDIFF describes invertible differences between a source and a target. These support the general idea of context-assisted encoding, not a guarantee for every memory or compression ratio. [1, 2]

Deliberately link the cart boundaries

Rick proposes using the tail of the preceding cart and the head of the following cart. These directions refer to the ordered record sequence. Encode explicit boundary links or overlaps, and retain the current cart's unique residual: whatever those boundaries cannot reconstruct. This makes useful context a design target, not an assumption that neighbors are automatically sufficient. Exactness still requires an unambiguous decoder and verified round trips.

A small exact example

Suppose cart A contains the exact ASCII text The lantern is blue. with no quotation marks or trailing newline. Cart B originally contains The lantern is green. under the same convention.

An illustrative encoding for B is: reference A; copy its first 15 bytes; append the six literal bytes green. The copied prefix includes the space after is. With A and these exact instructions, the decoder reconstructs B byte for byte. Without A, the reference alone cannot supply that prefix.

The example is a decoding rule, not an optimal compressor. For short strings, storing the reference and instructions may cost more than simply retaining B. Its purpose is to show where the supposedly missing information still lives.

Context is useful only when it constrains the answer

Carts before or after a record may reduce what needs to be stored, but proximity does not guarantee useful information. If two different originals remain compatible with the entire retained bundle, exact reconstruction is not guaranteed. A plausible completion, or a sentence with the same meaning, is not recovery of the original bytes.

The full train is the accounting unit

Measure the total stored bundle, not just the smallest cart. Count dependent neighbors, reference identifiers, indexes, dictionaries, independently decodable checkpoints and integrity metadata. If a particular model or model state is required for decoding, specify it and account for its storage or availability. Moving information into another cart does not make that information free.

Shared context can reduce redundancy, but cannot promise unlimited compression for arbitrary data. Exact recovery requires the decoder and all required side information to be available in the stated versions. Lossy summaries remain useful retrieval views; they do not become lossless merely because neighboring summaries sound consistent.

Make dependencies recoverable

Earlier and later carts can both be used if the decoding order is well defined. A later cart may already exist in an archive, but may not yet be available in a live stream. That distinction affects latency and when reconstruction is possible.

Avoid circular references with no independent starting point. Retain independently decodable checkpoints, explicit dependency links and provenance. Store integrity hashes to detect changed or damaged content, while remembering that a hash verifies a candidate rather than reconstructing the missing record. Decide how far a missing cart may disrupt the rest of the train.

A small verification test

1. Freeze a set of original records and define the target: exact bytes, exact text under a declared encoding, or semantic similarity. Do not report the third as the first.

2. Encode the records, then decode using only the permitted bundle and declared decoder. Keep a separate original copy solely for verification; it must not secretly supply information during decoding.

3. Compare reconstructed bytes directly with the originals. Repeat on held-out records that were not used to tune the method. Record total bundle size and reconstruction cost, including dependencies.

4. Remove, alter or reorder a neighboring cart, and test missing future context. The decoder should detect an unavailable or invalid dependency, recover from a retained checkpoint where possible, or report failure rather than silently inventing a replacement.

A successful test would establish exact reconstruction for the tested records and configuration. It would not by itself establish universal losslessness, perfect human-like recall or consciousness. The useful next question is which information each cart retains and which other carts its decoder is allowed to rely on.

Technical references

[1] P. Deutsch. DEFLATE Compressed Data Format Specification version 1.3. RFC 1951, sections 2 and 3.2. https://www.rfc-editor.org/rfc/rfc1951

[2] D. Korn et al. The VCDIFF Generic Differencing and Compression Data Format. RFC 3284, sections 1 and 3. https://www.rfc-editor.org/rfc/rfc3284